

Audsley's Optimal Priority Assignment: A Peer Review and Implementation

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Abstract—In this paper, we explore Audsley's approach to fixed priority assignment for task asynchronous task-set scheduling. We begin with a short discussion of familiar fixed-priority scheduling algorithms such as Rate Monotonic and Deadline Monotonic, followed by a description of when these methods are no longer optimal. We then explore how Audsley's algorithm for fixed-priority assignment works, and explain its optimality. Finally, we implement the three scheduling algorithms and run them on various task sets to compare schedulability.

Index Terms—Fixed priority, Audsley's algorithm, Task offset, Common release time

I. INTRODUCTION

There are many ways to rank the utility of scheduling algorithms, one of which is according to optimality with respect to meeting all deadlines for a given task set. Earliest Deadline First (EDF) is well-known to be optimal with respect to schedulability when executing periodic task sets on a uniprocessor system. However, there exist alternative scheduling algorithms which offer other advantages. For example, fixed priority scheduling algorithms such as Rate Monotonic (RM) and Deadline Monotonic (DM) often have simpler implementation and analysis, as well as improved predictability over EDF.

Unfortunately, when tasks are allowed to have arbitrary start times, the task set may no longer be optimal with RM and DM scheduling. It is also

Rate Monotonic (RM)

Deadline Monotonic (DM)

Related Work. Contributions.

- 1) First, we provide an overview of when RM and DM fail, and show some motivating examples of task sets which are schedulable, but produce infeasible schedules for RM and DM.
- 2) Next, we describe Audsley's approach and show that it is optimal and produces feasible schedules when RM and DM fail.
- 3) Finally, we implement Audsley's priority assignment scheme in Python, and evaluate its performance with respect to schedulability against RM and DM for characteristic task sets.

Organization. The remainder of this paper is organized as follows. We detail our motivating algorithms and task sets in Sec. II, describe our system model in Sec. III, explain Audsley's algorithm implementation and its results in Sec. IV, and conclude in Sec. V.

II. BACKGROUND & MOTIVATIONS

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TABLE I
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Table Head	Table Column Head		
	<i>Table column subhead</i>	<i>Subhead</i>	<i>Subhead</i>
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^aSample of a Table footnote.



Fig. 1. Example of a figure caption.

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ACKNOWLEDGMENT

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REFERENCES

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